



Weekly Report

- **Title:** Fortnightly Training Report
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- **Student ID:** 445002604
- **Program:** Bachelor of Computer Science
- **Week:** Week 3 and Week 4
- **Dates:** 1 Feb to 12 Feb 2026
- **Organization/Company Name:** Wadi Makkah Technology Company
- **Supervisor Name:** Mr. Khalid Abdulghani
- **Submission Date:** 12 Feb 2026

1. Introduction

This report aims to document the tasks and activities carried out during the third and fourth weeks of the cooperative training period. This phase focused on developing skills in building APIs using Node.js and Express.js and connecting them to databases, along with learning to use React in web development. A practical project combining the front-end and back-end was also developed, and enhancing teamwork skills, branch management, and merging on GitHub.

2. Tasks and Responsibilities

Week 3:

- **Calculator Project:** Performed a branch merge on GitHub to release the final version of the "Calculator" project. I merged the changes made by the Frontend and Backend teams, ensured code compatibility, and resolved potential conflicts. This process unified the teams' efforts and produced a stable, complete version of the project ready for deployment.
- **Frontend Development with React:** Completed a React course to strengthen frontend development skills, covering core concepts such as Components, State management, Hooks, and API integration.

- **Portfolio:** Developed a personal portfolio using React. The project emphasized reusable React components, Hooks to manage scroll events and dynamically update the Navbar, and State management to toggle project details interactively. A visually appealing and user-friendly interface was designed using CSS with interactive effects to enhance the user experience. This project improved my understanding of modern web interface design and interactivity.

- **React Presentation:** Delivered a presentation demonstrating how to convert a traditional portfolio website into a React-based portfolio. The presentation covered React components and state management to enhance user interaction, while explaining the benefits of React in terms of code reusability, project organization, and performance improvement. A practical demonstration was included to highlight the differences between the traditional version and the React-based version.

Week 4:

- **Backend API Development:** Developed a Backend API using Node.js and Express.js to receive numerical values from users and perform arithmetic operations. Two types of requests were handled: GET requests, which receive values via the URL and are accessed using req.query, and POST requests, which receive values from the request body and are handled using req.body. The result is returned in JSON format, demonstrating successful communication between the client and the server.

- **CRUD Task System Development:** A Backend API was developed connected to a SQLite database using the Sequelize library. The task focused on creating a Task Management system supporting basic CRUD operations. Async/await was used to ensure proper handling of asynchronous operations, and all routes were tested using Thunder Client to verify the API's functionality.

- **Full-Stack Task Management:** The Task Management API was integrated with a React-based user interface, allowing users to interact with tasks directly through the frontend. Users can add tasks, update their status to complete, and delete tasks. React state and the fetch API were used to manage data and communicate with the backend, reflecting changes in real-time, which enhanced the user experience and demonstrated smooth and organized interaction between the frontend and backend.

3. Skills and Knowledge Gained

- Developing interactive front-end interfaces using React.
 - Building back-end systems with Node.js and Express.
 - Designing APIs and handling GET and POST requests.
 - Integrating the front-end with the back-end using the Fetch API for real-time data updates.
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- Managing databases using SQLite and connecting them via Sequelize ORM.
 - Developing full-featured web applications with seamless component interaction.
 - Practical knowledge of React architecture and modern web development concepts.
 - Version control, branch management, and merging using Git and GitHub.
 - Gaining teamwork and collaborative skills through hands-on projects.
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4. Challenges Faced

- I faced difficulty connecting the frontend with the backend API, as each part was running on a different port, which caused issues with updating and displaying data in real-time.
 - When integrating the calculator frontend with the backend, some buttons did not work correctly due to errors in linking the JavaScript files in the merged code.
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5. Solutions and Improvements

- The port mismatch issue was resolved by installing and using the CORS library to allow the frontend to communicate with the backend on a different port, along with using React state to manage data and reflect changes in real-time.
 - A new branch was created for merging, the merged code was reviewed, JavaScript links were corrected, all button functionalities were tested, and the revised version was then merged into the main branch on GitHub.
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6. Conclusion

During these two weeks, I had a hands-on experience turning ideas into practical projects. I faced real challenges, devised practical solutions, and gained insights into how system components interact between the frontend and backend. This experience helped refine my technical skills and develop my ability to tackle challenges, turning mistakes into opportunities for learning and growth.

End of Report



